Paper IV – Recent Advances (MCh Urology, Summer 2018)

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IV-1. Radiology for solid renal masses.

Type: LAQ (25 marks)

Introduction

- Most renal masses are now found incidentally. Imaging is the **strongest predictor of malignancy**; size and **enhancement** are the two key variables.
- About 20% of solid enhancing cT1 masses are benign (oncocytoma, fat-poor AML (angiomyolipoma)).
- **Goals:** solid or cystic? enhancing? fat? subtype? stage and surgical anatomy.

A. CT (computed tomography): the standard test

Renal mass protocol (dedicated, thin slices):

Phase	Timing	Use
Non-contrast	–	Baseline HU (Hounsfield units), fat, calcification, haemorrhage
Corticomedullary	30–70 s	Hypervascular masses; arteries; clear cell vs other subtypes
Nephrographic	90–120 s	**Most sensitive** for detection (tumour enhances less than parenchyma); renal vein and IVC (inferior vena cava) thrombus
Excretory	7–10 min	Collecting system relation

Enhancement thresholds (rise from unenhanced value):

Rise	Meaning
≥20 HU	Enhancing: solid tumour
10–20 HU	Equivocal (MRI, magnetic resonance imaging, or CEUS)
<10 HU	Non-enhancing: cyst

- **Pitfall:** pseudoenhancement of small (<10–15 mm) cysts; dual-energy CT iodine maps help.
- **Macroscopic fat = AML:** fat ≤ −10 to −20 HU on unenhanced CT is diagnostic. Fat **with calcification** suggests RCC (renal cell carcinoma).
- **Fat-poor AML:** hyperdense on unenhanced CT, homogeneous, avid early enhancement and washout.

- **Subtype clues:** clear cell RCC enhances early, avidly and heterogeneously (necrosis); papillary RCC is homogeneous and **hypoenhancing**; chromophobe and oncocytoma intermediate, with a central scar. Overlap is large.

B. MRI

- **Indications:** contrast allergy, CKD (chronic kidney disease), equivocal CT enhancement, young patients, thrombus. Enhancement = signal rise ≥15–20%. Macrocyclic gadolinium carries very low NSF (nephrogenic systemic fibrosis) risk.

Feature	Clear cell	Papillary	Chromophobe / oncocytoma	AML
T2 signal	High, heterogeneous	**Low**, homogeneous (haemosiderin)	Central T2-bright scar	Fat signal
Enhancement	Avid (>200%)	Weak	Intermediate / avid	Avid
Diffusion (ADC, apparent diffusion coefficient)	High	Lowest	Intermediate	–
Chemical shift	**Signal drop** (intracellular lipid, ~59%)	No drop	–	**India-ink** artefact; loss on fat saturation

- **ccLS (clear cell likelihood score, 1–5):** structured MRI algorithm for solid cT1a masses using T2 signal, corticomedullary enhancement and microscopic fat, plus diffusion and **segmental enhancement inversion** (favours oncocytoma). Score 4–5 suggests clear cell RCC; 1–2 a non-clear-cell tumour. It guides biopsy versus surveillance.

C. Ultrasound and CEUS (contrast-enhanced ultrasound)

- Grey-scale: cyst versus solid; poor below 2 cm. AML is hyperechoic, but so are some small RCCs.
- **CEUS** (microbubbles, not nephrotoxic): shows enhancement; used in CKD or allergy, indeterminate lesions, ablation and surveillance.

D. Cystic masses: Bosniak 2019 (CT and MRI)

Class	Key feature	Malignancy / action
I	Simple, thin wall ≤2 mm, no septa	Benign; none
II	1–3 thin septa; homogeneous hyperdense (≥70 HU unenhanced)	Benign; none
IIF	Wall or septa 3 mm, or ≥4 thin septa	<10%; follow 6, 12 months, then yearly for 5 years
III	Wall or septa ≥4 mm, or irregular	~50%; surgery or surveillance
IV	**Enhancing nodule**	~90%; treat

E. Functional and molecular imaging

- **99mTc-sestamibi SPECT/CT** (single-photon emission CT): mitochondria-rich **oncocytoma** and hybrid oncocytic or chromophobe tumours take it up; clear cell RCC does not.
- **Girentuximab PET (positron emission tomography)** targets CA-IX (carbonic anhydrase IX) on clear cell RCC. **ZIRCON** (^{89}Zr-girentuximab, indeterminate cT1 masses): about 85% sensitivity and 87% specificity for clear cell RCC.
- **FDG-PET** (fluorodeoxyglucose PET): not recommended (low sensitivity). PSMA (prostate-specific membrane antigen) PET: experimental.

F. Renal mass biopsy

- **Indications:** result may change management; suspected lymphoma, metastasis or infection; before ablation or surveillance; metastatic disease. Not needed if surgery is planned regardless.
- Cores: PPV (positive predictive value) 98.8%; subtype accuracy ≥90%, grade 60–80%; non-diagnostic ~14%. Seeding almost nil. Oncocytoma cannot be confirmed on cores.

G. Staging and surgical planning

- **Venous thrombus:** CT sensitivity 78% (renal vein), 96% (IVC). MRI or multiplanar CT shows the cephalad tip; enhancing thrombus is tumour, non-enhancing is bland.

Mayo level	Extent	T stage
0	Renal vein	T3a
I	IVC ≤2 cm above renal vein	T3b
II	IVC below hepatic veins	T3b
III	Intrahepatic IVC, below diaphragm	T3b
IV	Above diaphragm or atrium	T3c

- Nodes ≥2 cm usually malignant. Chest CT; bone scan only if symptomatic.
- **Nephrometry:** R.E.N.A.L. (Radius, Exophytic/endophytic, Nearness to collecting system, Anterior/posterior, Location), PADUA, C-index: predict complexity and complications; guide partial versus radical nephrectomy.
- **After ablation:** persistent or new enhancement = residual or recurrence.
- **Recent:** radiomics and AI (artificial intelligence).

Diagram to draw: algorithm: incidental mass → renal-protocol CT/MRI → fat (AML) / cyst (Bosniak) / enhancing solid → biopsy or treat.

> **Key points to remember:**
> - Enhancement ≥20 HU = solid tumour; macroscopic fat = AML.
> - MRI: clear cell T2-bright with chemical-shift drop; papillary T2-dark and hypoenhancing; ccLS 1–5.
> - Sestamibi for oncocytoma; ^{89}Zr-girentuximab PET (ZIRCON) for clear cell.
> - Biopsy PPV ~99%, non-diagnostic ~14%.

IV-2. Diagnosis and management for Adrenal Carcinoma.

Type: LAQ (25 marks)

Introduction and epidemiology

- ACC (adrenocortical carcinoma): rare (0.7–2 per million per year), aggressive cortical tumour. **Bimodal** (children <10 years; adults 40–50 years); slight female excess.
- **Syndromes:** Li–Fraumeni (TP53), Beckwith–Wiedemann (IGF-2, insulin-like growth factor 2), Lynch, MEN-1 (multiple endocrine neoplasia 1), Carney complex, FAP (familial adenomatous polyposis). Surgery is the only cure.

Clinical features

- **Functional in 50–79% of adults** (90% of children):
  - Cortisol (commonest): rapidly progressive Cushing syndrome, often with virilisation.
  - Androgens: virilisation in women. **Oestrogen in a man (feminisation) suggests carcinoma.**
  - Aldosterone: rare.
- Non-functional: pain, fullness, weight loss, or incidental.

Hormonal work-up (ESE–ENSAT, European Society of Endocrinology–European Network for the Study of Adrenal Tumours)

Axis	Tests
Glucocorticoid	1 mg overnight dexamethasone suppression test (or 24-h urine or late-night salivary cortisol); ACTH (adrenocorticotropic hormone)
Sex steroids and precursors	DHEA-S (dehydroepiandrosterone sulphate), androstenedione, testosterone, 17-OH-progesterone, 11-deoxycortisol; oestradiol in men and postmenopausal women
Mineralocorticoid	Aldosterone:renin ratio if hypertension or hypokalaemia
Catecholamine	Plasma or urine **metanephrines**: must exclude phaeochromocytoma

- Why: confirms cortical origin, plans steroid cover, gives **tumour markers** for follow-up.

Imaging

- **Size:** mean 10–12 cm; >90% exceed 5 cm. Carcinoma in ~2% <4 cm, ~6% at 4–6 cm, ~25% >6 cm.
- **CT (computed tomography):** irregular, heterogeneous, necrosis, calcification; **unenhanced >10 HU (Hounsfield units)** (mean ~39 HU vs ~8 HU in adenoma); **slow washout** (absolute <60%, relative <40% at 15 min). Growth ~1.9 cm/year.
- **MRI (magnetic resonance imaging):** no chemical-shift signal loss; best for IVC (inferior vena cava) thrombus and invasion.
- **FDG-PET/CT** (fluorodeoxyglucose positron emission tomography): avid, but cannot separate ACC from metastasis.
- **Staging:** CT chest, abdomen, pelvis; bone or brain only if symptomatic.
- **No biopsy** if ACC is suspected (seeding, poor accuracy); only for unresectable or metastatic disease, after excluding phaeochromocytoma.

Pathology: Weiss score (≥3 of 9 = malignant)

Domain	Criteria
Nuclear / cytology	High nuclear grade; >5 mitoses per 50 high-power fields; atypical mitoses

Domain	Criteria
Architecture	≤25% clear cells; diffuse architecture; necrosis
Invasion	Venous; sinusoidal; capsular

- SF-1 (steroidogenic factor 1) confirms cortical origin. **Ki-67** and mitotic rate grade the tumour (low vs high grade).

ENSAT staging

Stage	Definition	5-year disease-specific survival (ENSAT 2009)
I	T1 (≤5 cm), N0 M0	~80%
II	T2 (>5 cm), N0 M0	~60%
III	T3 (periadrenal fat) or T4 (adjacent organ, venous thrombus) or N1	~50%
IV	Distant metastasis only	~13%

- ENSAT differs from older UICC (Union for International Cancer Control) staging by moving T4 and N1 without metastasis into stage III.

Management

1. Surgery (stages I–III)

- **Open transperitoneal adrenalectomy** for suspected ACC, especially >6 cm or invasive. Laparoscopy risks capsule rupture and peritoneal carcinomatosis; only for small organ-confined tumours by experts.
- **R0, intact capsule:** rupture gives dismal results. **En bloc** removal of involved kidney, liver, spleen, pancreas or IVC; IVC thrombectomy (bypass if needed).
- **Locoregional lymphadenectomy** (periadrenal fat, renal hilum, para-aortic or paracaval) improves staging and may reduce recurrence.
- High-volume surgeon (ESE–ENSAT: >6, preferably ≥20 adrenalectomies a year). Hydrocortisone cover if cortisol-secreting.

2. Adjuvant therapy

- **Mitotane** (adrenolytic DDT (dichlorodiphenyltrichloroethane) derivative) for **high risk**: stage III, R1/Rx, or Ki-67 >10% (ESE–ENSAT 2018), for at least 2 years.
  - Target blood level **14–20 mg/L**; needs **glucocorticoid replacement** (it destroys the cortex and raises cortisol-binding globulin).
  - Toxicity: gastrointestinal, neurological, hepatic, hypothyroidism, gynaecomastia.
- **ADIUVO** (RCT, randomised controlled trial; R0, stage I–III, Ki-67 ≤10%): no recurrence-free survival benefit. So observe low-risk patients.
- **Adjuvant radiotherapy** (tumour bed): R1 or Rx, or stage III. It cuts local recurrence but not overall survival.

3. Advanced or metastatic (stage IV)

- **EDP-M** (etoposide, doxorubicin, cisplatin + mitotane) is first line. **FIRM-ACT** (2012): versus streptozotocin–mitotane, response 23% vs 9% and PFS (progression-free survival) 5.0 vs 2.1 months; overall survival not significantly better.
- Mitotane alone for low-volume, slow disease.
- **Surgery:** resect oligometastases or late recurrence; debulk only if >90% removable.
- **Hormone control:** metyrapone, ketoconazole, mifepristone, etomidate. Radiotherapy for bone and brain metastases.

Follow-up

- CT chest, abdomen, pelvis and baseline-raised hormones **every 3 months for 2 years**, then 6-monthly to 5 years.

Prognosis

- Overall 5-year survival **20–47%**. Bad factors: stage, R1, high Ki-67 or mitoses, cortisol excess, age, size >12 cm.

Recent advances

- Molecular classes (CIMP, CpG island methylator phenotype) predict prognosis. Checkpoint inhibitors give modest responses; IGF-1R (IGF-1 receptor) inhibitor linsitinib failed (GALACCTIC). ADIUVO-2 tests mitotane ± platinum–etoposide in high risk.

Diagram to draw: algorithm: adrenal mass → hormones + CT (HU, washout) → suspected ACC → open R0 en bloc resection + LND (lymph node dissection) → risk-based adjuvant mitotane or radiotherapy.

> **Key points to remember:**
>
> - Feminisation in a man, >6 cm, >10 HU, poor washout: think ACC. Never biopsy before resection.
> - Weiss ≥3/9; ENSAT stage IV = distant metastasis only.
> - Open, R0, intact capsule, en bloc + lymphadenectomy.
> - Mitotane level 14–20 mg/L; ADIUVO (low risk: no benefit); FIRM-ACT: EDP-M first line.

IV-3a. Struvite stones.

Type: SAQ (10 marks)

Definition and epidemiology

- **Infection stones:** magnesium ammonium phosphate hexahydrate ($MgNH_4PO_4 \cdot 6H_2O$), usually mixed with **carbonate apatite**.
- 5–15% of stones (falling); female:male ~2:1; highest in India and Pakistan. Risk: neurogenic bladder, spinal injury, diversion, catheters, obstruction, diabetes, elderly.

Pathogenesis

```
Urease-producing bacteria
  Urea + H2O -> 2NH3 + CO2
  NH3 -> NH4+ (urine pH > 7.2)
  CO2 -> carbonate; phosphate -> PO4(3-)
  + Mg -> struvite;  + Ca -> carbonate apatite
```

- Sterile urine has no urease, so infection is essential. Ammonia damages the GAG (glycosaminoglycan) layer; bacteria live inside the stone in a **biofilm**, so antibiotics cannot sterilise a retained stone.
- **Organisms: Proteus mirabilis** (commonest), Klebsiella, Pseudomonas, Providencia, Staphylococcus, **Ureaplasma** (easily missed). E. coli is rarely urease-positive.

Clinical features

- Recurrent UTI (urinary tract infection), fever, flank pain, haematuria; often few symptoms. Grow fast into **staghorn** calculi. Untreated: pyonephrosis, XGP (xanthogranulomatous pyelonephritis), sepsis, renal loss.

Evaluation

- Urine pH >7; culture (urine and stone cultures often differ). NCCT (non-contrast computed tomography): burden and anatomy; DMSA (dimercaptosuccinic acid) or renogram for split function. Stone analysis; 24-h urine (many have hypercalciuria or hypocitraturia).

Management

- **Complete stone clearance** is the goal: each fragment is an infected nidus.
- Culture-specific antibiotics before surgery; drain an obstructed infected kidney first.
- **PCNL (percutaneous nephrolithotomy): first line** for staghorns (AUA, American Urological Association). Add flexible nephroscopy, ECIRS (endoscopic combined intrarenal surgery) or second look. SWL (shock wave lithotripsy) or RIRS (retrograde intrarenal surgery) for small stones only.
- Anatrophic nephrolithotomy rarely; nephrectomy for a poorly functioning kidney.
- **Prevention:** surveillance cultures, suppressive antibiotics, fluids, correct stasis or metabolic defects.
- **AHA (acetohydroxamic acid)** 250 mg three times daily (urease inhibitor), only when surgery is exhausted. Side effects in up to 30% (haemolytic anaemia, DVT (deep vein thrombosis)); avoid if creatinine >2.5 mg/dL or pregnancy.
- **Also know:** hemiacidrin irrigation (historical); matrix stones (radiolucent, Proteus).

Diagram to draw: urease cascade; staghorn with PCNL tract.

Key points to remember:

- Urease → ammonia → pH >7.2 → struvite + carbonate apatite.
- Proteus mirabilis commonest; Ureaplasma easily missed.
- Complete clearance with PCNL; AHA only when surgery is not possible.

IV-3b. Klinefelter's syndrome.

Type: SAQ (10 marks)

Definition and genetics

- **47,XXY**: commonest sex chromosome aneuploidy and **commonest genetic cause of male infertility**; 1 in 500–1000 male births.
- Non-mosaic in **80–90%**; the rest mosaic **46,XY/47,XXY** (milder, more sperm) or 48,XXXY variants. Extra X from non-disjunction (maternal age).
- Underdiagnosed: most never diagnosed; increasingly detected on NIPT (non-invasive prenatal testing).

Pathophysiology

- Seminiferous tubules **hyalinise** progressively from puberty; germ cells are lost. Leydig cells fail, so testosterone falls. Puberty starts on time, then stalls.

Clinical features

Age	Features
Infant	Usually normal; micropenis, cryptorchidism, hypospadias
Adolescent/adult	**Small firm testes (<3.5 cm, 1–4 mL)**, gynaecomastia, tall stature with long legs (only ~30%), sparse hair, female fat distribution
Neurodevelopment	Language, learning and attention difficulties

Hormones and tests

- **High FSH (follicle-stimulating hormone) and LH (luteinising hormone), low or low-normal testosterone, high oestradiol** (raised oestradiol:testosterone ratio). Inhibin B low.
- Semen: **azoospermia** in most (~8% have ejaculated sperm).
- **Karyotype** confirms (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine, guideline: karyotype all men with non-obstructive azoospermia or severe oligozoospermia).

Management

- **Fertility: micro-TESE** (microdissection testicular sperm extraction) + ICSI (intracytoplasmic sperm injection). Sperm found in **~40–50%** (68% in one series); offspring are usually chromosomally normal.
- Spermatogenesis declines with age, so retrieve early (late adolescence or young adulthood). Some give clomiphene, anastrozole or hCG (human chorionic gonadotropin) beforehand.
- **TRT (testosterone replacement therapy):** for hypogonadism (bone, muscle, mood, libido). It **suppresses spermatogenesis**, so offer sperm retrieval or banking **before** starting.
- Gynaecomastia: reduction mammoplasty if persistent. Speech, educational and psychological support.

Associated risks

- Osteoporosis, metabolic syndrome, diabetes, VTE (venous thromboembolism), autoimmune disease.
- **Mediastinal (extragonadal) germ cell tumours**; male breast cancer (~1% lifetime).

Diagram to draw: hypothalamic–pituitary–testis axis showing high FSH and LH, low testosterone.

Key points to remember:

- 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
- Small firm testes, high FSH and LH, azoospermia.
- Micro-TESE ~50% success; retrieve sperm before testosterone.
- Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

IV-3c. Brain dead committee.

Type: SAQ (10 marks)

Assumed meaning: the Board of Medical Experts that certifies brain-stem death under Indian law.

Legal basis

- **THOA (Transplantation of Human Organs Act) 1994** legalised brain-stem death in India; **amended 2011** (Transplantation of Human Organs and Tissues Act); **Rules 2014**.
- Brain-stem death: irreversible loss of consciousness and capacity to breathe.

Composition of the Board (Rules 2014)

Member	Role
1. Registered medical practitioner **in charge of the hospital**	Hospital authority
2. **Independent specialist**, nominated by the in-charge from a panel approved by the Appropriate Authority	Independent check
3. **Neurologist or neurosurgeon** from the approved panel (if unavailable: surgeon, physician, anaesthetist or intensivist)	Expert
4. Medical practitioner **treating the patient**	Clinical history

- **No member may belong to the transplant team.**

Certification procedure

- **Preconditions:** known irreversible cause, apnoeic coma on ventilator; exclude sedatives, relaxants, hypothermia, metabolic or endocrine causes, shock.
- **Two sets of tests, at least 6 hours apart**, recorded on **Form 10**.

Brainstem reflex (all absent)	Method	Cranial nerves
Pupillary	Fixed pupils, no light reaction	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic	Turn head: eyes move with head	VIII, III, VI
Oculovestibular	Ice-cold water in each ear: no eye movement	VIII, III, VI
Motor in cranial nerve area	Supraorbital pressure: no grimace	V, VII
Gag and cough	Pharynx and tracheal suction: no response	IX, X

- **Apnoea test:** pre-oxygenate with 100% oxygen, disconnect ventilator with oxygen insufflation; **no breathing effort at $PaCO_2$ (arterial CO_2) ≥60 mmHg**. Abort if hypotension, desaturation or arrhythmia.

Linked duties

- **Mandatory request:** ICU (intensive care unit) staff must inform relatives of the option to donate; a **transplant coordinator** counsels; next of kin give consent.
- **NOTTO** (National Organ and Tissue Transplant Organisation), with regional and state bodies (ROTTO, SOTTO: regional and state organ and tissue transplant organisations), allocates organs and keeps the registry.

Diagram to draw: flowchart: coma → preconditions → test 1 → 6 h → test 2 → Form 10 → consent → NOTTO.

> **Key points to remember:**
>
> - 4-member Board; no transplant-team member.
> - Two tests ≥6 h apart; Form 10.
> - Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.

IV-3d. Chronic non bacterial prostatitis.

Type: SAQ (10 marks)

Definition and classification

- Now called **CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)**, NIH (National Institutes of Health) category III: pelvic pain for **≥3 of the last 6 months** without uropathogenic infection. 90–95% of prostatitis.

NIH category	Description
I	Acute bacterial prostatitis
II	Chronic bacterial prostatitis (recurrent same-organism UTI)
IIIA	CP/CPPS, **inflammatory** (leucocytes in EPS (expressed prostatic secretion), VB3 (post-massage urine) or semen)
IIIB	CP/CPPS, **non-inflammatory**
IV	Asymptomatic inflammatory

- IIIA vs IIIB has no treatment value.

Aetiology

- Interplay of past infection, neurogenic inflammation, **pelvic floor muscle dysfunction**, autoimmunity, psychological stress and **central sensitisation**. The prostate may not be the pain source.

Clinical features

- Perineal (commonest), testicular, suprapubic or penile pain; ejaculatory pain; LUTS (lower urinary tract symptoms); sexual dysfunction.

Evaluation (diagnosis of exclusion)

- **NIH-CPSI (Chronic Prostatitis Symptom Index):** pain, urinary and quality-of-life domains (0–43).
- DRE (digital rectal examination): prostate tender in fewer than half; palpate the pelvic floor for **trigger points** (tender in ~half).
- Urinalysis and culture; **two-glass test** (pre- and post-massage). Cytology if irritative symptoms or haematuria. PVR (post-void residual), flow.
- Selected: cystoscopy, urodynamics.
- **UPOINT phenotyping: U**rinary, **P**sychosocial, **O**rgan-specific, **I**nfection, **N**eurological/systemic, **T**enderness of muscles.

Management: multimodal, phenotype-directed (monotherapy trials fail)

Domain	Therapy
Urinary	**α-blockers** in recent-onset, α-blocker-naive men (<1 year); antimuscarinics
Psychosocial	CBT (cognitive behavioural therapy), stress reduction
Organ (prostate)	**Quercetin**, **pollen extract** (small RCTs, randomised controlled trials), 5-ARI (5α-reductase inhibitor) if BPH (benign prostatic hyperplasia)
Infection	Quinolone or tetracycline for ≥6 weeks if treatment-naive and <1 year (EAU, European Association of Urology), or positive VB3; no repeated courses
Neurological	Pregabalin, amitriptyline, duloxetine

Domain	Therapy
Tenderness	**Pelvic floor physiotherapy**, myofascial trigger-point release, biofeedback

- **Others: acupuncture** (beat sham in a large RCT), PTNS (percutaneous tibial nerve stimulation), shockwave therapy, aerobic exercise; NSAIDs (non-steroidal anti-inflammatory drugs) only in combination.
- **Not recommended:** TURP (transurethral resection of the prostate), radical prostatectomy.

Diagram to draw: UPOINT snowflake with a therapy for each domain.

> **Key points to remember:**
> - NIH III: pain ≥3 of 6 months, no infection; IIIA inflammatory, IIIB not.
> - NIH-CPSI to score; UPOINT to phenotype.
> - Multimodal therapy; α-blocker plus antibiotic only early.

IV-3e. Semen analysis nomenclature.

Type: SAQ (10 marks)

Collection (WHO, World Health Organization, 2021, 6th edition)

- Masturbation into a clean wide-mouthed non-toxic container (special condom if needed); complete sample.
- Abstinence **2–7 days** (evidence: 1–2 days best). Keep at 20–37 °C; analyse within **1 hour**, after liquefaction (~30 min).
- At least **two samples, 2–3 weeks apart**, because of variability.

Lower reference limits (5th centile, fertile men)

Parameter	Lower limit
Volume	**1.4 mL**
Concentration	**16 million/mL**
Total count	**39 million/ejaculate**
Total motility	**42%**
Progressive motility	**30%**
Normal forms (strict)	**4%**
Vitality	**54%**
Peroxidase-positive leucocytes	<1 million/mL

- These are reference values, not fertility cut-offs. The 6th edition grades motility as progressive, non-progressive or immotile.

Nomenclature

Term	Meaning
Normozoospermia	All parameters at or above reference
Oligozoospermia	Total count (or concentration) below reference
Asthenozoospermia	Progressive motility below reference
Teratozoospermia	Normal forms below reference
OAT (oligoasthenoteratozoospermia)	All three abnormal
Azoospermia	No sperm, **including centrifuged pellet**
Cryptozoospermia	Sperm absent in fresh smear, present only after centrifugation
Aspermia	No ejaculate (dry orgasm)
Hypospermia	Volume below reference
Necrozoospermia	Low vitality (many dead sperm)
Leucocytospermia	Peroxidase-positive white cells above 1 million/mL
Haemospermia	Blood in ejaculate
Globozoospermia	Round-headed sperm without acrosome

Notes

- **Low volume or aspermia:** post-ejaculatory urine for retrograde ejaculation; TRUS (transrectal ultrasound) for EDO (ejaculatory duct obstruction).

- **Vitality:** eosin–nigrosin, or **HOS (hypo-osmotic swelling)** test, which keeps sperm usable for ICSI (intracytoplasmic sperm injection).
- Only azoospermia is definitive; values overlap between fertile and infertile men.
- **Also know:** Kruger strict morphology; DNA fragmentation; antisperm antibodies (MAR, mixed antiglobulin reaction); semen pH no longer recommended.

Diagram to draw: normal sperm (head, midpiece, tail) beside a WHO reference box.

> **Key points to remember:**
> - WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
> - Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
> - Two samples 2–3 weeks apart; abstinence 2–7 days.

IV-3f. Combined Androgen Blockade.

Type: SAQ (10 marks)

Definition and rationale

- **CAB (combined androgen blockade)**, also called maximal or total androgen blockade: castration (orchidectomy or LHRH (luteinising hormone-releasing hormone) agonist) **plus a non-steroidal antiandrogen**.
- **Rationale (Labrie):** castration removes ~90–95% of testosterone, but **adrenal androgens** (DHEA, dehydroepiandrosterone; androstenedione) are converted to DHT (dihydrotestosterone) inside the prostate. An antiandrogen blocks these at the AR (androgen receptor).

Antiandrogens

Drug	Dose	Signature toxicity
Flutamide	250 mg three times daily	Diarrhoea, hepatotoxicity
Nilutamide	300 mg then 150 mg daily	Delayed dark adaptation, pneumonitis
Bicalutamide	**50 mg daily** (with LHRH agent)	Best tolerated; gynaecomastia
Cyproterone (steroidal)	100 mg two to three times daily	Cardiovascular, hepatic; not used for CAB now

- Class effects: gynaecomastia, hepatotoxicity, pneumonitis.

Evidence

- **PCTCG (Prostate Cancer Trialists' Collaborative Group) meta-analysis, 2000** (27 trials, ~8,000 men): with non-steroidal antiandrogens, 5-year survival improved by only **~2–3%** (borderline significance). Cyproterone gave no benefit (slightly worse).
- So the benefit is small, with more cost and toxicity. **AUA (American Urological Association): long-term CAB not recommended**.

Current uses

1. **Flare prevention:** an LHRH agonist raises LH (luteinising hormone) and testosterone for 10–20 days, risking cord compression, pain or obstruction. Start antiandrogen **with or before** the agonist for **21–28 days**. (Antagonists such as degarelix or relugolix cause no flare.)
2. **Antiandrogen withdrawal:** on progression, stopping bicalutamide gives PSA (prostate-specific antigen) fall >50% in 15–30% (mutant AR made the drug an agonist).

Modern intensification (replaces CAB in mHSPC, metastatic hormone-sensitive prostate cancer)

Trial	Addition to ADT (androgen deprivation therapy)	Result
LATITUDE, STAMPEDE (arm G)	Abiraterone + prednisone	OS (overall survival) HR (hazard ratio) 0.62 / 0.63
ENZAMET	Enzalutamide (vs ADT + older antiandrogen)	OS HR 0.67
ARCHES	Enzalutamide	Radiographic PFS (progression-free survival) HR 0.39
TITAN	Apalutamide	OS HR 0.67

Trial	Addition to ADT (androgen deprivation therapy)	Result
PEACE-1, ARASENS	**Triplet**: + docetaxel + abiraterone or darolutamide	OS benefit (ARASENS HR 0.68)

Diagram to draw: testis and adrenal androgen sources, with LHRH agonist and antiandrogen sites of action.

Key points to remember:

- CAB = castration + non-steroidal antiandrogen (adrenal androgens).
- PCTCG: ~2–3% 5-year survival gain only.
- Main use now: flare cover for 21–28 days.
- ARPIs (androgen receptor pathway inhibitors) or triplets now standard in mHSPC.

Rapid Revision Sheet – Paper IV

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

IV-1. Radiology for solid renal masses. (LAQ)

- Enhancement ≥20 HU = solid tumour; macroscopic fat = AML.
- MRI: clear cell T2-bright with chemical-shift drop; papillary T2-dark and hypoenhancing; ccLS 1–5.
- Sestamibi for oncocytoma; ^{89}Zr-girentuximab PET (ZIRCON) for clear cell.
- Biopsy PPV ~99%, non-diagnostic ~14%.

IV-2. Diagnosis and management for Adrenal Carcinoma. (LAQ)

- Feminisation in a man, >6 cm, >10 HU, poor washout: think ACC. Never biopsy before resection.
- Weiss ≥3/9; ENSAT stage IV = distant metastasis only.
- Open, R0, intact capsule, en bloc + lymphadenectomy.
- Mitotane level 14–20 mg/L; ADIUVO (low risk: no benefit); FIRM-ACT: EDP-M first line.

IV-3a. Struvite stones. (SAQ)

- Urease → ammonia → pH >7.2 → struvite + carbonate apatite.
- Proteus mirabilis commonest; Ureaplasma easily missed.
- Complete clearance with PCNL; AHA only when surgery is not possible.

IV-3b. Klinefelter's syndrome. (SAQ)

- 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
- Small firm testes, high FSH and LH, azoospermia.
- Micro-TESE ~50% success; retrieve sperm before testosterone.
- Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

IV-3c. Brain dead committee. (SAQ)

- 4-member Board; no transplant-team member.
- Two tests ≥6 h apart; Form 10.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.

IV-3d. Chronic non bacterial prostatitis. (SAQ)

- NIH III: pain ≥3 of 6 months, no infection; IIIA inflammatory, IIIB not.
- NIH-CPSI to score; UPOINT to phenotype.
- Multimodal therapy; α-blocker plus antibiotic only early.

IV-3e. Semen analysis nomenclature. (SAQ)

- WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
- Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
- Two samples 2–3 weeks apart; abstinence 2–7 days.

IV-3f. Combined Androgen Blockade. (SAQ)

- CAB = castration + non-steroidal antiandrogen (adrenal androgens).
- PCTCG: ~2–3% 5-year survival gain only.
- Main use now: flare cover for 21–28 days.
- ARPIs (androgen receptor pathway inhibitors) or triplets now standard in mHSPC.